
VVU HBK Mk-II

Terminal-as-a-Service (TaaS) Commercial Framework Specification

Phase 2 — Power & Thermal Architecture + Commercial Model

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Author: VVU (Venture Vision Ubuntu)

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1. Executive Summary

The Terminal-as-a-Service (TaaS) model represents a fundamental strategic pivot from traditional CapEx-based hardware procurement toward a Non-Recourse OpEx Treatment framework for municipal water infrastructure. Under conventional procurement paradigms, municipalities are compelled to allocate substantial upfront capital for hardware acquisition, installation, and commissioning — creating an immediate balance sheet liability with no guarantee of operational performance or data yield.

The TaaS framework eliminates this structural inefficiency by transferring the entire hardware lifecycle — from manufacturing through deployment, maintenance, and eventual decommissioning — under VVU's operational umbrella. The core value propositions are:

CapEx to OpEx Pivot: Municipalities receive the HBK Mk-II Hydro-Gateway Assembly as a fully operational service unit without assuming ownership liability. The subscription-based OpEx structure allows deployment within existing operational budgets, bypassing prohibitive CapEx approval cycles that delay critical infrastructure modernization by years.

Balance Sheet De-risking: The hardware remains on VVU's balance sheet, not the client's. Depreciation risk, obsolescence exposure, and residual value uncertainty are entirely absorbed by VVU. The municipality's balance sheet carries only a predictable operational expense line item.

100% Equity Retention: VVU retains absolute ownership of all core technology, platform IP, and Terminal hardware. No client, partner, or third party may acquire ownership stakes. The hardware never becomes a liability on the client's balance sheet, and VVU never dilutes its control over the terminal's operational parameters.

Sole Data Sovereignty: All data collected by VVU Terminals belongs exclusively to VVU. The Observation Vector (O_t) — comprising Pressure, Flow, Acoustic, and Temperature dimensions — constitutes proprietary intelligence that no third party may access, replicate, or redistribute without VVU's explicit written authorization.

2. TaaS Core Pillars

2.1 100% Equity Retention

VVU maintains absolute ownership of every HBK Mk-II Hydro-Gateway Assembly deployed under the TaaS framework. This is not a lease arrangement with transfer provisions, nor a financed purchase with a vesting schedule — it is an uncompromised equity position that persists from manufacturing through deployment, operation, and eventual decommissioning or redeployment.

Non-Transferable Rights: No client, partner, or third party may acquire ownership stakes in the deployed hardware. The commercial relationship remains a pure service contract — the client pays for intelligence delivery, not asset acquisition.

Asset Security: Because VVU retains full ownership, the hardware cannot be seized, lien-attached, or otherwise encumbered by municipal financial distress. In the event of client default or contract termination, VVU exercises its unilateral recovery right.

Capital Efficiency: The 100% equity position grants VVU the exclusive right to retrieve and redeploy HBK Mk-II assets across successive service engagements — transforming each hardware unit from a single-deployment cost center into a multi-lifecycle revenue generator.

2.2 Sole Data Sovereignty

Under the TaaS framework, VVU holds exclusive and non-shareable sovereignty over all data generated by the deployed HBK Mk-II terminal. The Observation Vector (O_t) comprises four primary measurement dimensions: **P** (Pressure), **F** (Flow), **A** (Acoustic signature), and **T** (Temperature). No client, academic partner, or third-party vendor may access, replicate, or redistribute this data without VVU's explicit written authorization.

The sovereignty framework is enforced through **Data-Audit Inseparability**: every Observation Vector packet is cryptographically bound to its generating terminal via SHA-256 hash chaining. This creates an immutable audit trail ensuring data provenance, integrity, and non-tampering. The Bayesian inference algorithms, probabilistic models, and localization methodologies that transform raw O_t data into actionable intelligence are VVU's proprietary intellectual property.

2.3 InfrastructureRight Abstraction

The TaaS framework introduces the InfrastructureRight Abstraction — a four-pillar resource interface that transforms physical infrastructure access into digital rights within a pooled resource interface. The four pillars are:

Water: Hydraulic access and flow measurement rights — the municipality subscribes to a Water Right that guarantees a defined level of hydraulic intelligence delivery.

Energy: Power supply and backup entitlements — guaranteed power continuity for the terminal's operational availability.

Compute: Bayesian inference processing allocation — the probabilistic leak-localization processing capacity delivered as a service.

Storage: Telemetry data retention and archival capacity — secure, sovereign data persistence with guaranteed retention periods.

This abstraction decouples the client's infrastructure requirements from the specific hardware implementation. The municipality does not purchase "a pressure sensor" or "a flow meter" — it subscribes to a Water Right that guarantees a defined level of hydraulic intelligence delivery. If VVU upgrades the sensor array or reconfigures the inference pipeline, the client's service level remains constant — the InfrastructureRight is hardware-agnostic.

3. Hydro-Gateway Assembly — 11 Integrated Components

The HBK Mk-II Hydro-Gateway Assembly comprises 11 integrated components, each positioned within a precise XYZ coordinate framework that defines the physical architecture of the terminal. The coordinate system establishes the spatial relationships necessary for structural integrity, hydraulic coupling, and Bayesian inference processing.

Component	X (mm)	Y (mm)	Z (mm)	Functional Role
Pressure Pipe	0	0	750	Nodal Head Estimation
South Datum Skid	0	-560	40	Base structural support
North Datum Skid	0	+560	40	Base structural support
Left Service Rack	-720	-240	290	Lateral mounting
Right Service Rack	+720	+240	290	Lateral mounting
Edge Control Cabinet	0	+400	400	Bayesian inference
Power Backup Module	0	-400	400	Power redundancy
Inlet Meter Pod	-750	-160	930	Primary flow measurement
Outlet Meter Pod	+750	+160	930	Secondary flow measurement
Telemetry Mast	0	0	1290	Data transmission
Top Height Beacon	+100	0	1465	Site identification

The datum skids (North and South) anchor the assembly to the installation surface, while the service racks provide lateral stability. The Edge Control Cabinet houses the Bayesian inference engine that transforms raw sensor observations into probabilistic leak-localization outputs. The dual meter pods generate the differential flow measurements essential for the Observation Vector, and the Telemetry Mast transmits encrypted data packets to VVU's central processing infrastructure.

4. Revenue Split — 60/30/10

All TaaS subscription revenue is distributed according to a fixed 60/30/10 allocation model that balances operational sustainability, growth investment, and risk governance. This split is contractually fixed and cannot be modified without triparty agreement.

Allocation	Percentage	Scope	Description
Operational Baseline	60%	Hardware + Operations	Hardware amortization, server costs, HBK inference runtime
Growth & R&D	30%	Software + Research	Software refinement, Bayesian algorithm improvement, scaling, academic collaboration

Allocation	Percentage	Scope	Description
Risk & Compliance	10%	Audit + Regulatory	ProofBridge-Liner audit chain, regulatory alignment, SRS parameter verification

4.1 Operational Baseline (60%)

The 60% Operational Baseline allocation covers the essential cost structure of terminal deployment and operation: hardware amortization across the terminal's projected lifecycle, server infrastructure costs for the HBK inference engine runtime, field service operations including calibration and maintenance, and the operational overhead of maintaining 24/7 terminal availability. This allocation ensures that the core service delivery remains financially sustainable independent of R&D; or compliance budget fluctuations.

4.2 Growth & R&D; (30%)

The 30% Growth & R&D; allocation funds the continuous improvement pipeline: Bayesian algorithm refinement and next-generation inference models, software platform development and scaling, academic collaboration funding for independent validation research, and next-generation terminal development. This allocation is the engine of the TaaS model's competitive moat — it ensures that the terminal's intelligence capabilities improve continuously rather than stagnating at the initial deployment level.

4.3 Risk & Compliance (10%)

The 10% Risk & Compliance allocation funds the governance infrastructure: ProofBridge-Liner audit chain operations, regulatory alignment and compliance documentation, SRS parameter verification, and the independent audit functions that validate the terminal's performance claims. This allocation protects both VVU and the municipal client from operational risk by ensuring that all performance claims are independently verified and documented.

5. Verification Gates — VR1 to VR5

The Zero Fabrication Mandate is enforced through a five-stage verification gate system that progressively validates the terminal from geometric specification through field acceptance. Each gate represents a mandatory checkpoint that must be passed before the next stage can proceed. No gate may be skipped, combined, or conditionally waived.

Gate	Designation	Scope	Validation Criteria
VR1	Geometry Verification	CMM data, nominal diameter	All 11 component XYZ positions verified; nominal diameter 114.3 mm confirmed
VR2	Material Verification	Mill test reports, NDT	Material certificates, corrosion resistance testing, structural strength validation
VR3	Assembly Verification	Torque logs, route card sign-offs	Inter-component connectivity, hydraulic seal integrity, electrical bonding verification

Gate	Designation	Scope	Validation Criteria
VR4	Functional Verification	FAT, sensor calibration	Factory Acceptance Testing under simulated conditions, Bayesian inference output validation
VR5	Field Acceptance	SAT, municipal commissioning	Site Acceptance Testing at deployment site, live hydraulic conditions, telemetry confirmation

The verification gates establish a linear, non-parallel validation path that ensures every dimension of the terminal's specification is independently confirmed. VR1 and VR2 are prerequisites for Tranche 1 budget release, ensuring that no manufacturing capital is deployed before the terminal's physical and material specifications have been independently validated.

6. Zero Fabrication Mandate — 32 SRS Parameters

The Zero Fabrication Mandate prohibits any production manufacturing of the HBK Mk-II Hydro-Gateway Assembly until all 32 System Requirements Specification (SRS) parameters have been independently verified against their respective verification gates. The 32 parameters are distributed across the five verification gates: 3 Geometry (VR1), 3 Material (VR2), 2 Assembly (VR3), 15 Functional (VR4), and 9 Field (VR5).

#	Parameter	Category	Target	Gate	Status
1	Nominal Diameter	Geometry	114.3 mm	VR1	Pending
2	Wall Thickness	Geometry	6.02 mm	VR1	Pending
3	Overall Length	Geometry	750 mm	VR1	Pending
4	Material Grade	Material	API 5L X42	VR2	Pending
5	Yield Strength	Material	≥ 290 MPa	VR2	Pending
6	Corrosion Allowance	Material	≤ 3.0 mm	VR2	Pending
7	Bolt Torque	Assembly	85 ± 5 Nm	VR3	Pending
8	Seal Integrity	Assembly	0 leak points	VR3	Pending
9	Pressure Range	Functional	0 — 16 bar	VR4	Pending
10	Flow Range	Functional	0 — 500 m ³ /h	VR4	Pending
11	Acoustic Bandwidth	Functional	0.1 — 500 Hz	VR4	Pending
12	Temperature Range	Functional	-10 — +60 °C	VR4	Pending
13	Bayesian Inference Latency	Functional	≤ 200 ms	VR4	Pending
14	Leak Localization Radius	Functional	≤ 500 m	VR4	Pending
15	False Positive Rate	Functional	$\leq 5\%$	VR4	Pending
16	Sensor Sampling Rate	Functional	≥ 100 Hz	VR4	Pending
17	Data Throughput	Functional	≥ 1.2 Mbps	VR4	Pending
18	Power Consumption	Functional	≤ 45 W	VR4	Pending

#	Parameter	Category	Target	Gate	Status
19	Backup Duration	Functional	≥ 4 hours	VR4	Pending
20	EMC Compliance	Functional	IEC 61000	VR4	Pending
21	Surge Tolerance	Functional	≤ 20 bar	VR4	Pending
22	Wave Celerity	Functional	1400 m/s	VR4	Pending
23	Inference Accuracy	Functional	≥ 95%	VR4	Pending
24	Site Commissioning	Field	SAT passed	VR5	Pending
25	Telemetry Uplink	Field	≥ 99.5% uptime	VR5	Pending
26	Environmental Sealing	Field	IP67	VR5	Pending
27	Vibration Resistance	Field	IEC 60068	VR5	Pending
28	UV Resistance	Field	ISO 4892	VR5	Pending
29	Corrosion Field Test	Field	ISO 9223	VR5	Pending
30	Municipal Integration	Field	SCADA compatible	VR5	Pending
31	Regulatory Compliance	Field	SANS 241	VR5	Pending
32	Operational Handover	Field	Sign-off complete	VR5	Pending

No production fabrication may commence until all 32 parameters have achieved "Verified" status through their respective gate validations. The Prototyping Exemptions clause permits limited experimental validation builds (1-3 units) for laboratory testing and calibration, funded from the 30% R&D; allocation — not from the Tranche 1 tooling deposit.

7. SLA Metrics

7.1 Leak Localization

The primary SLA metric for the TaaS framework is leak localization accuracy, measured as the reduction in search uncertainty relative to conventional detection methods. The HBK Mk-II terminal, operating under its Bayesian inference engine, commits to achieving a **95% reduction in search uncertainty** compared to traditional acoustic correlation methods. This translates to a **maximum search radius of 500 meters** for any detected leak event.

The accuracy target is achieved through the convergence of three technical capabilities: the Observation Vector's four-dimensional sensor data (P, F, A, T), the Bayesian inference engine's probabilistic spatial mapping, and the SHA-256 hash-chained data integrity that ensures no sensor drift or data corruption degrades the localization output.

7.2 False Positive Rate

False-positive events represent the most costly failure mode in leak detection operations. The TaaS SLA framework addresses this through a strict **False-Positive Rate (FPR) target of ≤ 5%**, enforced via

Poisson-Gaussian mixture distributions in the Bayesian inference engine. The Poisson component models the discrete event frequency (how many leak events occur in a given observation window), while the Gaussian component models the continuous measurement noise (sensor signal variability under normal operating conditions). The mixture distribution enables the inference engine to distinguish between genuine hydraulic anomalies and statistical noise fluctuations with high confidence.

The 5% FPR target is validated against historical pipeline data from comparable municipal networks and is subject to continuous recalibration as the terminal accumulates site-specific operational data. Each false-positive event is logged, analyzed, and fed back into the inference engine's training dataset — ensuring that the FPR metric improves over the deployment lifecycle.

7.3 Information Density

The Information Density (ID) metric is the ultimate efficiency indicator for the TaaS framework, defined as:

ID = Information Gained / Deployment Cost

This ratio measures the operational intelligence yield per unit of financial investment. Unlike raw accuracy metrics that measure detection performance in isolation, Information Density contextualizes performance within the commercial framework — ensuring that the intelligence delivery is not merely accurate but economically efficient. The ID metric provides the Municipal Client with a quantifiable return-on-investment indicator and provides VVU with an internal efficiency benchmark.

SLA Metric	Target	Method	Governance
Leak Localization	≤ 500m radius (95% reduction)	Bayesian inference spatial probability mapping	30% R&D; allocation for algorithm refinement
False Positive Rate	≤ 5%	Poisson-Gaussian mixture distribution analysis	Continuous recalibration against site-specific data
Information Density	ID = Info Gained / Deploy Cost	Ratio of intelligence yield to subscription investment	Internal efficiency benchmark driving R&D; priority

8. Vendor Financing

8.1 Tranche 1 — Budget Lock

The TaaS vendor financing framework establishes a structured capital release mechanism that enforces the Zero Fabrication Mandate at the financial level. The **Tranche 1 Budget Lock of R812,490.00** is a ring-fenced tooling deposit that covers the production mold fabrication and the KCL (Kinematic Calibration Library) parametric library development. This deposit is released only upon successful completion of Verification Gate VR1 (Geometry) and VR2 (Material) — ensuring that no manufacturing capital is deployed before the terminal's physical and material specifications have been independently validated.

Category	Amount (ZAR)	Tranche	Release Condition
Hardware	R340,000	Tranche 1	VR1 + VR2 completion
Software	R185,000	Tranche 1	VR1 + VR2 completion
Academic Collaboration	R127,490	Tranche 1	VR1 + VR2 completion
Municipal Pilot	R160,000	Tranche 2	VR4 (FAT) completion
Operational Scaling	R500,000	Tranche 3	VR5 (SAT) completion + first deployment

8.2 Tranche Structure

The three-tranche structure ensures that capital is released only against verified milestones. Tranche 1 (R812,490) covers tooling and initial development, released upon VR1/VR2 completion. Tranche 2 (R160,000) funds the municipal pilot deployment, released upon VR4 (Factory Acceptance Testing) completion. Tranche 3 (R500,000) funds operational scaling, released upon VR5 (Site Acceptance Testing) completion and the first successful deployment. This progressive release mechanism ensures that VVU's manufacturing investment is preceded by validated specifications.

9. Triparty SRS Delegation Addendum — "Three Keys"

The Triparty SRS Delegation Addendum establishes a three-key governance structure for the System Requirements Specification verification process. Each keyholder holds an independent, non-transferable validation authority over the SRS parameters. No verification gate may be declared passed without the affirmative sign-off of all three keyholders. The "Three Keys" framework ensures that no single party can unilaterally advance the terminal through the verification pipeline — creating a structural safeguard against specification shortcuts, verification conflicts of interest, and governance failures.

Keyholder	Key Designation	Role	Validation Authority
VVU	Master Key	Technology Owner	Full authority over SRS parameter definitions, design specifications, and technical architecture. VVU defines what the terminal must achieve.
Academic Partner	Validation Key	Independent Validator	Independent verification authority over SRS parameter compliance. The Academic Partner confirms that VVU's specifications are met through objective, reproducible testing.
Municipal Authority	Operational Key	Operational Authority	Operational acceptance authority over field deployment conditions. The Municipal Authority confirms that the terminal meets the operational requirements of the deployment environment.

9.1 Key Interdependence

The three keys are structurally interdependent: no single key can override the others. The Master Key (VVU) defines what must be verified, the Validation Key (Academic Partner) confirms that verification is genuine, and the Operational Key (Municipal Authority) confirms that the verified specification meets the deployment environment's requirements. This separation of concerns — definition, validation, and acceptance — ensures that the SRS verification process is both technically rigorous and operationally relevant.

The Three Keys framework also provides the governance foundation for the ProofBridge-Liner audit chain. Every verification gate sign-off is recorded as a triparty-signed audit entry, creating an immutable record of who validated what, when, and on what basis. This audit trail is maintained independently of any single keyholder and is accessible to all three parties for dispute resolution.

10. Default & Asset Recovery Provisions

The default and asset recovery framework is the ultimate enforcement mechanism of VVU's 100% equity retention principle. In the event of client default — whether through subscription payment failure, contract violation, or operational non-compliance — VVU exercises its unilateral right to recover and redeploy HBK Mk-II assets. This right is absolute and non-negotiable.

Provision	Timeline	Description
Terminal Return	Within 30 days	Hardware returned to VVU upon default confirmation. VVU dispatches field recovery team to decommission and extract the terminal assembly.
Data Extraction	Within 14 days	VVU extracts all stored data from the terminal. Municipal copy provided within 14 days of extraction. VVU retains original data under sole sovereignty.
IP Preservation	Immediate	VVU retains all IP, model weights, and inference code. No client, partner, or third party may retain copies of VVU's proprietary algorithms or trained models.
ProofBridge-Liner Audit	Activated on default	Independent audit chain activated. All terminal data, verification records, and operational logs are sealed and preserved for independent review.
Escalation Protocol	After 60 days	Payment default exceeding 60 days triggers reduced-function mode. Terminal operates in degraded state with limited intelligence output until payment is remediated.

10.1 Recovery Procedure

Upon default event confirmation, VVU issues a formal Recovery Notice to the Municipal Client with a 30-day compliance window. If the default is not remediated within this window, VVU dispatches a field recovery team to decommission and extract the terminal assembly. The recovered asset enters VVU's refurbishment pipeline: components are inspected, recalibrated, and upgraded as necessary before redeployment at a subsequent service location. The recovery cost is absorbed by VVU's 10% Risk & Compliance allocation — ensuring that asset recovery does not impose additional financial burden on the Operational Baseline or R&D budget lines.

10.2 Reduced-Function Mode

The escalation protocol provides a graduated response to payment default. After 60 days of non-payment, the terminal transitions to reduced-function mode: the Bayesian inference engine operates with diminished accuracy, the telemetry uplink frequency is reduced, and the Information Density metric is correspondingly degraded. This mode is not a punitive measure — it is a cost-alignment mechanism that reduces VVU's operational expenditure on the terminal while the default is being resolved. Full operational capability is restored immediately upon payment remediation, with no re-verification required.

10.3 Asset Lifecycle Continuity

The recovery mechanism transforms the TaaS model from a rental arrangement into a true service lifecycle: the hardware is never "sold and forgotten" but remains a perpetually managed commercial asset that generates revenue across multiple deployment cycles. The unilateral recovery right ensures that VVU's capital investment in each HBK Mk-II unit is protected regardless of individual client outcomes — de-risking the overall portfolio and enabling confident investment in next-generation terminal development.

VVU-TaaS-CFS-2025-001 — END OF SPECIFICATION